



Finding Profession

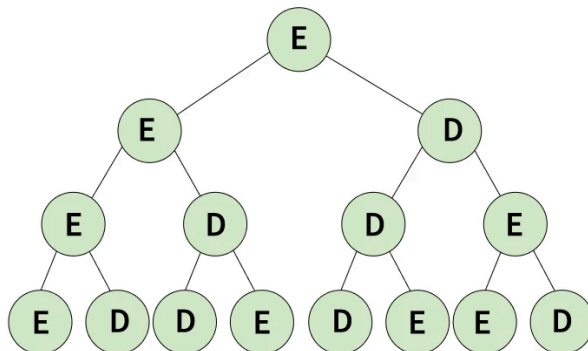


Difficulty: **Medium** Accuracy: **30.3%** Submissions: **32K+**
Points: **4** Average Time: **20m**

Consider a special family of **Engineers** and **Doctors** with following rules :

- Everybody has two children.
- First child of an Engineer is an Engineer and second child is a Doctor.
- First child of an Doctor is Doctor and second child is an Engineer.
- All generations of Doctors and Engineers start with Engineer.

The first few levels of the family tree are shown below :



Given **level** and position (**pos**) of a person in above ancestor tree, find profession of the person.

```
1 class Solution:
2     def profession(self, level, pos):
3         flips = bin(pos - 1).count('1')
4         return "Doctor" if flips % 2 else "Engineer"
```



Examples:**Input:** level = 4, pos = 2**Output:** Doctor**Explanation:** It is shown in the tree given in question.**Input:** level = 3, pos = 4**Output:** Engineer**Explanation:** Already given in the tree in question.[Custom Input](#)[Compile & Run](#)[Submit](#)